Recursion Problems

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Problem Solving Seminar

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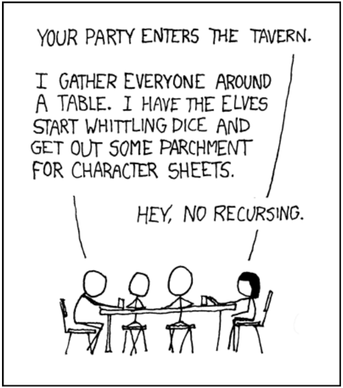

Recursive definitions involves two things:

- a base case – where the thing is defined explicitly
- a recursive step – a rule which relates the next case to previous cases

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$$F_2 = 1, \quad F_3 = 2, \quad F_4 = 3, \quad F_5 = 5, \quad F_6 = 8, \ldots\ldots$$

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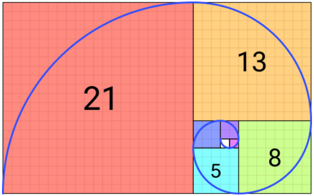

Putnam 1990 A1

Let $T_0 = 2$, $T_1 = 3$, $T_2 = 6$, and for $n \geq 3$,

$$T_n = (n+4)T_{n-1} - 4nT_{n-2} + (4n-8)T_{n-3}.$$

The first few terms are

$$2, 3, 6, 14, 40, 152, 784, 5168, 40576, 363392$$

Find, with proof, a formula for T_n of the form $T_n = A_n + B_n$, where (A_n) and (B_n) are well-known sequences.

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Find, with proof, a formula for T_n of the form $T_n = A_n + B_n$, where (A_n) and (B_n) are well-known sequences.
Hint: compare to the first few terms of the sequence $n!$

$$1, 1, 2, 6, 24, 120, 720, 5040, 40320, 362880$$

Linear with constant coefficients

Special case:

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$$u_4 = (a^3 + 2ab)u_1 + (a^2 b + b^2)u_0$$

Linear with constant coefficients

Linearize:

$$\begin{bmatrix} u_{n+1} \\ u_{n+2} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ b & a \end{bmatrix} \begin{bmatrix} u_n \\ u_{n+1} \end{bmatrix}$$

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$$\vec{u}_n = A^n\vec{u}_0$$

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$$\begin{bmatrix} F_n \\ F_{n+1} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}^n \begin{bmatrix} F_0 \\ F_1 \end{bmatrix}$$

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$$\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}^4 = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$$

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What about $\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}^{10}$?

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Let

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

Observation:

$$A^2 - A^1 - I = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

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Polynomial long division:

$$\frac{x^{10}}{x^2 - x - 1} = x^8 + x^7 + 2x^6 + 3x^5 + 5x^4 + 8x^3 + 13x^2 + 21x + 34 + \frac{55x + 34}{x^2 - x - 1}$$

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$$A^{10} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} + 55\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} + 34\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 34 & 55 \\ 55 & 89 \end{bmatrix}$$

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$$c = \frac{\varphi_+^{10} - \varphi_-^{10}}{\varphi_+ - \varphi_-} = 55, \quad d = \frac{\varphi_+^{10} - \varphi_-^{10}}{\varphi_+ - \varphi_-} = 34$$

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$$A^n = \frac{\varphi_+^n - \varphi_-^n}{\varphi_+ - \varphi_-} A + \frac{\varphi_+^{n-1} - \varphi_-^{n-1}}{\varphi_+ - \varphi_-} I$$

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$$F_n = \frac{\varphi_+^n - \varphi_-^n}{\varphi_+ - \varphi_-}.$$

Practice again

Suppose that $u_0 = 0$, $u_1 = 1$ and

$$u_{n+2} = 4u_{n+1} - 4u_n.$$

Determine the value of u_{16}.

Putnam 2015 A2

Suppose that $a_0 = 1$, $a_1 = 2$, and

$$a_n = 4a_{n-1} - a_{n-2}, \quad n \geq 2.$$

Determine an odd prime divisor of a_{2015}.

Generalization

Suppose that u_0, u_1, and u_2 are three real numbers. Define recursively

$$u_{n+2} = \frac{u_{n+1} + u_n + u_{n-1}}{3}.$$

Determine the limit of u_n.

Putnam 1993 A2

Let z_0 and z_1 be real numbers. Consider the sequence of real numbers defined recursively by

$$z_n^2 - z_{n+1}z_{n-1} = 1, \quad n \geq 1.$$

Show that there is a constant $a \in \mathbb{R}$ with

$$z_{n+1} + z_{n-1} = az_n, \quad n \geq 1$$

Can you find an explicit expression for z_n?